A First Course in Abstract Algebra

Section 20: Fermat's and Euler's Theorems

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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20 Fermat's and Euler's Theorems

Exercise 20.1: A generator modulo 7

The nonzero residues form a multiplicative group of order 6. Successive powers of 3, with reduction modulo 7 after every multiplication, are

$$3^1, 3^2, \ldots, 3^6 \equiv 3, 2, 6, 4, 5, 1 \pmod 7.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.2: A generator modulo 11

The nonzero residues form a multiplicative group of order 10. Successive powers of 2, with reduction modulo 11 after every multiplication, are

$$2^1, 2^2, \ldots, 2^{10} \equiv 2, 4, 8, 5, 10, 9, 7, 3, 6, 1 \pmod{11}.$$

Every nonzero residue appears once. Thus $\boxed{2}$ generates the entire multiplicative group.

Exercise 20.3: A generator modulo 17

The nonzero residues form a multiplicative group of order 16. Successive powers of 3, with reduction modulo 17 after every multiplication, are

$$3^1, 3^2, \ldots, 3^{16} \equiv 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1 \pmod{17}.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.4: A large power modulo twenty-three

Fermat's theorem applies because 23 is prime and does not divide 3. It gives $3^{22} \equiv 1 \pmod{23}$. Since $47 = 2 \cdot 22 + 3$,

$$3^{47} \equiv 3^3 = 27 \equiv \boxed{4} \pmod{23}.$$

Exercise 20.5: A large power modulo seven

Reduce the base: $37 \equiv 2 \pmod 7$. Fermat gives $2^6 \equiv 1 \pmod 7$, and $49 = 8 \cdot 6 + 1$. Therefore $37^{49} \equiv 2^{49} \equiv \boxed{2} \pmod 7$.

Exercise 20.6: A power whose exponent is itself a power

Since 19 is prime, $2^{18} \equiv 1 \pmod{19}$, so the outer exponent may be reduced modulo 18. The powers 2^k modulo 18 for $k = 1, \ldots, 6$ are $2, 4, 8, 16, 14, 10$, and $2^{k+6} \equiv 2^k$ for $k \geq 1$, because $2(2^6 - 1) = 126$ is divisible by 18. Thus $2^{17} \equiv 2^5 \equiv 14 \pmod{18}$. Now

$$2^{14} = 2^{10}2^4 \equiv 17 \cdot 16 = 272 \equiv 6 \pmod{19},$$

where $2^{10} = 1024 \equiv 17$. Consequently $2^{(2^{17})} + 1 \equiv \boxed{7} \pmod{19}$.

Exercise 20.7: Euler phi values through thirty

The value $\varphi(n)$ counts integers from 1 through n relatively prime to n; in particular $\varphi(1) = 1$. For computation, if the distinct prime divisors of n are p, use $\varphi(n) = n\prod_{p\mid n}(1 - 1/p)$. The complete table is

n	1	2	3	4	5	6	7	8	9	10
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4
n	11	12	13	14	15	16	17	18	19	20
$\varphi(n)$	10	4	12	6	8	8	16	6	18	8
n	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	12	10	22	8	20	12	18	12	28	8

Exercise 20.8: Euler phi for a prime square

Among $1, \ldots, p^2$, the integers not relatively prime to p^2 are exactly the p multiples $p, 2p, \ldots, p^2$. Removing them leaves $\boxed{\varphi(p^2) = p^2 - p = p(p-1)}$.

Exercise 20.9: Euler phi for a product of primes

If $p \neq q$, count the integers up to pq divisible by neither prime. There are q multiples of p, p multiples of q, and one multiple of both; inclusion–exclusion gives

$$\boxed{\varphi(pq) = pq - p - q + 1 = (p-1)(q-1)}.$$

If the primes are equal, the answer instead is $\varphi(p^2) = p(p-1)$ from Exercise 20.8. Distinctness cannot be silently assumed.

Exercise 20.10: Euler reduction modulo twenty-four

The base 7 is relatively prime to 24, and $\varphi(24) = 8$. Euler's theorem gives $7^8 \equiv 1 \pmod{24}$. Since $1000 = 125 \cdot 8$, the required remainder is $\boxed{1}$. One can also check directly that $7^2 = 49 \equiv 1 \pmod{24}$.

Exercise 20.11: A congruence with two residue solutions

The gcd is $\gcd(2, 4) = 2$, which divides 6. Divide the congruence by 2, including the modulus: $x \equiv 3 \pmod{2}$. Thus all integer solutions are $x = 1 + 2k$ with $k \in \mathbb{Z}$; the two incongruent solutions modulo 4 are $\boxed{1, 3}$.

Exercise 20.12: An invertible coefficient modulo fifteen

Reduce 22 modulo 15 to get $7x \equiv 5$. Since $7^{-1} = 13$ modulo 15, multiplication gives $x \equiv 65 \equiv 5 \pmod{15}$. Hence $\boxed{x = 5 + 15k, k \in \mathbb{Z}}$, one residue solution modulo 15.

Exercise 20.13: A congruence failing the gcd condition

Here $d = \gcd(36, 24) = 12$, but 12 does not divide 15. Every multiple $36x$ is divisible by 12, whereas $15 + 24k$ never is. Thus there are $\boxed{\text{no solutions}}$.

Exercise 20.14: Three residue solutions modulo twenty-four

Since $\gcd(45, 24) = 3$ divides 15, divide by 3 to obtain $15x \equiv 5 \pmod 8$, or $7x \equiv 5$. The inverse of 7 modulo 8 is 7, so $x \equiv 35 \equiv 3 \pmod 8$. Thus all integer solutions are $x = 3 + 8k$, and modulo 24 they are $\boxed{3, 11, 19}$.

Exercise 20.15: A second failure of the gcd condition

The coefficient 39 and modulus 9 have gcd 3. Since $3 \nmid 125$, the congruence has $\boxed{\text{no solutions}}$. Equivalently, its left side is zero modulo 3 but its right side is 2.

Exercise 20.16: One solution modulo nine

Reduce to $5x \equiv 8 \pmod 9$. Since $5 \cdot 2 \equiv 1$, multiply by 2: $x \equiv 16 \equiv 7 \pmod 9$. Thus $\boxed{x = 7 + 9k, k \in \mathbb{Z}}$.

Exercise 20.17: Five solutions modulo sixty-five

Here $\gcd(155, 65) = 5$, which divides 75. Divide by 5 to obtain $31x \equiv 15 \pmod{13}$, hence $5x \equiv 2$. Since $5^{-1} = 8$ modulo 13, $x \equiv 16 \equiv 3 \pmod{13}$. All integer solutions are $x = 3 + 13k$; the five residues modulo 65 are $\boxed{3, 16, 29, 42, 55}$.

Exercise 20.18: Thirteen solutions modulo one hundred thirty

The gcd is $\gcd(39, 130) = 13$, and $13 \mid 52$. Divide to obtain $3x \equiv 4 \pmod{10}$. Multiplying by $3^{-1} = 7$ gives $x \equiv 28 \equiv 8 \pmod{10}$. Hence all integer solutions are $\boxed{x = 8 + 10k}$, with $k \in \mathbb{Z}$; representatives modulo 130 are $8, 18, \ldots, 128$, thirteen in total.

Exercise 20.19: A factorial just below Wilson's factorial

Wilson's theorem gives $(p-1)! \equiv -1 \pmod p$. Since $(p-1)! = (p-1)(p-2)! \equiv -(p-2)!$, we obtain $\boxed{(p-2)! \equiv 1 \pmod p}$. For $p \geq 3$ the remainder is therefore 1.

Exercise 20.20: A factorial modulo thirty-seven

Write $36! = 36 \cdot 35 \cdot 34!$. Wilson's theorem and reduction of the last factors give $(-1)(-2)34! \equiv -1 \pmod{37}$, so $2 \cdot 34! \equiv -1$. The inverse of 2 is 19 modulo 37, and $-19 \equiv \boxed{18}$.

Exercise 20.21: A factorial modulo fifty-three

Wilson gives $52! \equiv -1 \pmod{53}$. Since $52! = 52 \cdot 51 \cdot 50 \cdot 49!$, we have $(-1)(-2)(-3)49! \equiv -1$. Thus $6 \cdot 49! \equiv 1$. The inverse of 6 modulo 53 is 9, because $6 \cdot 9 = 54 \equiv 1$. The remainder is $\boxed{9}$.

Exercise 20.22: A factorial modulo twenty-nine

Wilson gives $28! \equiv -1 \pmod{29}$. The four final factors are $28, 27, 26, 25 \equiv -1, -2, -3, -4$, whose product is 24. Hence $24 \cdot 24! \equiv -1$. Since $24 \cdot 23 = 552 \equiv 1$. Therefore $24! \equiv -23 \equiv \boxed{6} \pmod{29}$.

Exercise 20.23: True or false: Euler and congruences

(a) False: if $p \mid a$, then $a^{p-1} \equiv 0$, not 1.

(b) True: this is Fermat's theorem with its necessary nonzero-residue hypothesis.

(c) True: $\varphi(n)$ counts a subset of n integers, including $\varphi(1) = 1$.

(d) False as stated for all positive n: at $n = 1$, $\varphi(1) = 1 > 0 = n - 1$. It is true for $n \geq 2$, since n itself is then not coprime to n.

(e) True for $n \geq 2$, the usual nonzero-unity modular rings: the units are exactly residues with gcd 1 with n. If one admits $\mathbb{Z}_1$, the literal wording "positive integers less than n" omits its sole residue, so that case requires a separate convention.

(f) True: if u, v are units, $(uv)^{-1} = v^{-1}u^{-1}$.

(g) False: if ab is a unit in a commutative ring, $a[b(ab)^{-1}] = 1$ and similarly b is a unit; thus a product of two nonunits cannot be a unit.

(h) True by the same argument: a unit product would force the nonunit factor to be invertible.

(i) False: $0x \equiv 1 \pmod{p}$ has no solution.

(j) True: divide by d to obtain one solution modulo m/d, which lifts to exactly d distinct residue classes modulo m.

Exercise 20.24: The four units modulo twelve

The units are $1, 5, 7, 11$. Their table is

$\cdot$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

.

Every nonidentity element has order 2, so this is not the cyclic group of order 4. It is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$, for example via $1 \mapsto (0,0)$, $5 \mapsto (1,0)$, $7 \mapsto (0,1)$, $11 \mapsto (1,1)$.

Exercise 20.25: Synopsis of Fermat's theorem

For nonzero $a \in \mathbb{Z}_p$, Lagrange's theorem in the multiplicative group of the $p-1$ nonzero residues gives $a^{p-1} = 1$, which is precisely the desired congruence.

Exercise 20.26: Synopsis of Euler's theorem

The unit represented by an integer coprime to n lies in the multiplicative unit group of $\mathbb{Z}_n$, whose order is $\varphi(n)$, so Lagrange's theorem gives $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Exercise 20.27: Self-inverse nonzero residues

A self-inverse element satisfies $x^2 = 1$, hence $(x-1)(x+1) = 0$. Since $\mathbb{Z}_p$ is a field, one factor must vanish. Therefore $x = 1$ or $x = -1 = p-1$, and both do square to 1. For $p = 2$, these names describe the same element; for odd p, they are distinct.

Exercise 20.28: Wilson's congruence

For odd prime p, pair each nonzero residue other than 1 and -1 with its distinct multiplicative inverse. Every pair contributes product 1 to $(p-1)!$. By Exercise 20.27 the only unpaired factors are 1 and -1, giving $(p-1)! \equiv -1 \pmod{p}$. For $p = 2$, $1! = 1 \equiv -1 \pmod{2}$ directly.

For completeness, if an integer $n > 1$ satisfies this congruence, every prime divisor q of n must equal n: otherwise $q \leq n-1$ divides $(n-1)!$ and n, contradicting $(n-1)! \equiv -1 \pmod{q}$. Thus the converse also follows.

Exercise 20.29: A universal divisor of a polynomial value

Each prime p in $\{2, 3, 7, 13, 19, 37\}$ has $p - 1 \mid 36$. If $p \mid n$, then $n^{37} - n \equiv 0 \pmod{p}$. Otherwise Fermat gives $n^{36} \equiv 1 \pmod{p}$, again yielding $n^{37} - n = n(n^{36} - 1) \equiv 0$. These primes are distinct, so their product divides the difference:

$$2 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37 = \boxed{383838}.$$

Exercise 20.30: A larger universal divisor

Also $5 - 1 = 4$ divides 36, so the same Fermat argument shows $5 \mid n^{37} - n$ for every integer n. Since 5 is coprime to 383838, the larger number

$$\boxed{5 \cdot 383838 = 1919190}$$

divides every such difference.